

TENDERMIND

Tender Bid Agents Architecture Specification

An implementation-ready definition of agent roles, shared state, handoffs, risk synthesis, compliance controls, approval gates, and event-driven recalculation for AI-assisted tender bidding.

DOCUMENT agents.md specification

SYSTEM TenderMind

ARCHITECTURE Multi-agent tender intelligence + bid orchestration

STATUS Product / engineering draft

Core principle: every material conclusion is traceable to tender evidence, company memory, approved assumptions, or validated upstream agent output.

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1. System Purpose and Operating Principles

TenderMind is a multi-agent tender intelligence and bid-management system. When a tender, RFP, or RFQ is uploaded, the system does not immediately ask agents to generate a proposal.

The system first converts the source documents into a structured Tender Workspace containing requirements, clauses, BOQ items, deadlines, eligibility criteria, technical requirements, commercial terms, mandatory documents, and unresolved questions. Specialist agents then operate on this shared state.

What the system is designed to do

- Understand the tender before drafting.
- Determine whether the company should bid.
- Identify capability, eligibility, compliance, cost, schedule, legal, and commercial gaps.
- Calculate the realistic cost of execution.
- Prepare a compliant response with traceability.
- Surface decisions requiring human approval.
- Maintain institutional memory across bids.

Core operating principle

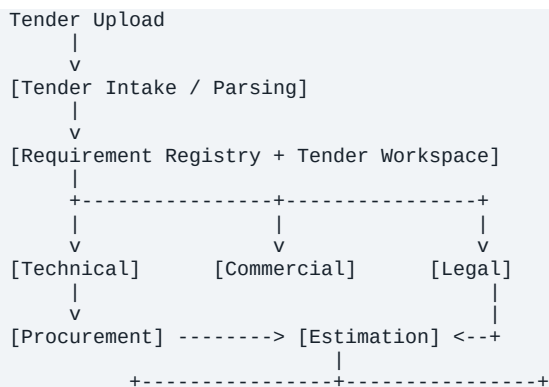
No invented tender facts: Agents may only rely on the tender, Company Knowledge Base, historical bid/project data, vendor or subcontractor quotes, user-approved assumptions, or another agent's validated output.

```
source:
  document_id:
  page:
  section:
  clause:
confidence:
assumption:
requires_human_review:
```

Unknown is a valid result: If information is unavailable or unsupported, return UNKNOWN rather than hallucinating a value.

2. System Architecture and Shared Workspace

Agent dependency flow





Shared Tender Workspace

All agents read and write against a single structured state object. This prevents each agent from reconstructing the tender independently and enables downstream invalidation when upstream facts change.

```

tender:
  tender_id:
  title:
  issuing_authority:
  tender_type:
  submission_deadline:
  clarification_deadline:
  pre_bid_meeting:
  bid_validity:
  estimated_contract_value:
  currency:
  submission_method:

requirements: []
technical_requirements: []
commercial_terms: []
legal_clauses: []
eligibility_requirements: []
boq_items: []
mandatory_documents: []
company_matches: []
company_gaps: []
vendor_quotes: []
assumptions: []
clarifications: []
risks: []
decisions: []
agent_outputs: {}

compliance_status:
  total_requirements:
  compliant:
  partial:
  non_compliant:
  unknown:

bid_status:
  stage:
  completion_percentage:
  blockers: []
  
```

Requirement object

```

requirement_id: REQ-001
category:
  technical | commercial | legal | financial | eligibility | submission

requirement:
  "Bidder must demonstrate five years of similar project experience."

source:
  document:
  page:
  section:
  clause:

mandatory: true
response_required: true
  
```

```

company_evidence: []

status:
  compliant | partial | non_compliant | unknown

responsible_agent:
risk_if_unmet:
clarification_required: false
notes:

```

3. Tender Intake / Parsing Agent

ROLE Convert raw tender documents into structured tender intelligence before specialist agents begin analysis.

INPUTS

- Tender PDF / RFP / RFQ documents
- BOQ spreadsheets
- drawings and technical appendices
- GCC / SCC and draft contract
- corrigenda / addenda
- submission instructions

RESPONSIBILITIES

- Extract tender metadata and all deadlines.
- Parse technical scope and requirements.
- Extract BOQ lines and quantities.
- Identify eligibility and qualification criteria.
- Extract mandatory forms and submission documents.
- Extract contractual and commercial clauses.
- Identify securities, guarantees, penalties, payment terms, and evaluation methodology.
- Preserve source location for every extracted requirement.

Outputs

```

tender_name:
buyer:
deadline:
contract_duration:
estimated_value:
submission_method:
bid_security:
performance_security:
evaluation_method:

clarification_questions: []

```

Guardrail: This agent performs extraction and classification, not qualification, pricing, legal interpretation, or bid decisions.

4. Technical / Engineering Agent

ROLE Determine how the company would actually execute the tender scope.

INPUTS

- Requirement Registry
- scope of work and technical specifications
- drawings

- Company Knowledge Base
- previous project methodologies
- resource and equipment capabilities

Technical compliance output

```
requirement:
company_capability:
status:
evidence:
gap:
recommended_response:
confidence:
```

Responsibilities

- Build execution methodology and work sequence.
- Define resource, manpower, and equipment plans.
- Map milestones, dependencies, testing, commissioning, and QA/QC obligations.
- Flag unavailable equipment, missing certifications, unrealistic schedules, unsupported specifications, and specialist subcontractor needs.
- Generate technical clarification questions where requirements are ambiguous.

Outputs

- Technical Compliance Matrix
- Technical Methodology
- Execution Plan
- Resource Plan
- Equipment Plan
- Technical Assumptions
- Technical Risks
- Technical Clarification Questions

Must not: Invent equipment ownership or certifications; approve commercial deviations; determine final cost; or accept contractual risk.

5. Procurement Agent

ROLE Determine the realistic external cost and availability of materials, equipment, vendors, and subcontractors.

Input-to-output schema

```
item:
quantity:
required_date:

vendors:
- vendor:
unit_rate:
lead_time:
validity:
payment_terms:
warranty:
compliance_status:

recommended_vendor:
```

```
selected_rate:
procurement_risk:
confidence:
```

Outputs

- Vendor Comparison Sheet
- Selected Procurement Rates
- Supplier Compliance Review
- Long Lead Item List
- Procurement Risks
- Rate Validity Flags

Always flag: Expired quotes, sole-source suppliers, unusually low prices, FX exposure, lead-time risk, and vendor qualification failures.

6. Estimation / Costing Agent

ROLE Calculate the true expected cost of delivering the tender. This agent behaves like an estimator, not a proposal writer.

Cost model

```
Material
+ Labor
+ Equipment
+ Subcontracting
+ Logistics
+ Site Costs
+ Direct Overhead
+ Allocated Indirect Overhead
+ Risk Contingency
= Expected Cost

Expected Cost
+ Target Margin
= Bid Price
```

BOQ output schema

```
boq_item:
description:
quantity:
unit:
material_cost:
labor_cost:
equipment_cost:
subcontractor_cost:
logistics_cost:
overhead:
contingency:
total_cost:
markup:
bid_rate:
bid_value:
data_sources: []
assumptions: []
confidence:
risk_flags: []
```

Outputs

- Priced BOQ
- Rate Analysis
- Cost Summary

- Margin Summary
- Cash-Flow Estimate
- Cost Assumptions
- Pricing Sensitivity
- Missing Cost Inputs

Conflict detection example: Technical requires three cranes for 12 months, while the cost model includes two cranes for eight months. This must become a Bid Manager conflict, not a silent correction.

7. Commercial / Contracts Agent

ROLE Evaluate the financial and contractual business terms of the tender.

Review areas

- Payment terms and advance payment
- retention and liquidated damages
- performance guarantees and bid security
- warranty obligations
- price escalation / price variation
- tax responsibilities
- foreign-exchange exposure
- invoice approval and change orders
- termination provisions

```
clause:
tender_position:
company_standard:
deviation:
financial_impact:
risk_level:
recommendation:
negotiation_position:
source:
```

Outputs

- Commercial Deviations Register
- Commercial Compliance Statement
- Cash-Flow Exposure
- Working Capital Impact
- Negotiation Items
- Commercial Risk Flags

8. Legal / Lawyer Agent

ROLE Identify legal exposure, eligibility failure, and unacceptable contractual obligations. The agent supports legal review but does not replace qualified counsel.

Review areas

- Liability caps, consequential damages, indemnities, and third-party claims.

- Governing law, arbitration venue, court jurisdiction, and dispute procedures.
- Licenses, permits, local registration, JV requirements, domestic content, and labor regulations.
- Mandatory legal qualification and bidder eligibility.

```
issue_id:
clause:
issue:
risk_type:
severity:
legal_exposure:
company_position:
recommended_action:
requires_external_counsel:
source:
```

Automatic critical flags

- UNLIMITED LIABILITY
- UNLIMITED INDEMNITY
- MANDATORY REQUIREMENT NOT MET
- UNACCEPTABLE JURISDICTION
- MISSING LICENSE
- JOINT-VENTURE REQUIREMENT
- BLACKLISTING DECLARATION ISSUE

Escalation: Critical legal issues automatically route to the Risk Matrix and Bid Manager.

9. Risk Matrix Agent

ROLE Act as the central risk synthesizer. It consumes validated risks produced by specialist agents and turns them into a unified risk model.

Implementation distinction: This is a consolidator, not a content generator. Use synthesis/conflict prompts, not drafting prompts.

```
risk_id:
description:
source_agent:
category:
likelihood: 1 | 2 | 3 | 4 | 5
impact: 1 | 2 | 3 | 4 | 5
risk_score:
financial_exposure:
schedule_exposure:
contractual_exposure:
mitigation:
residual_risk:
owner:
requires_leadership_decision:
pricing_contingency:
source:
```

Scoring

Risk Score = Likelihood x Impact

1-4	Low
5-9	Moderate
10-16	High
17-25	Critical

Outputs

- Master Risk Register

- Top 10 Bid Risks
- Risk Heat Map Data
- Risk-adjusted Contingency Recommendation
- Leadership Decision Requests
- Bid Blockers

Rule: The Risk Matrix may aggregate and prioritize; it must not silently override specialist conclusions. Material disagreement becomes a conflict.

10. Business Development / Bid Manager Agent

ROLE Act as the orchestrator of the entire bid: coordinate agents, track progress, resolve dependencies, surface decisions, and assemble the leadership review package.

Implementation distinction: Like the Risk Matrix, the Bid Manager is primarily a coordinator and synthesizer rather than a free-form content generator.

Progress tracking state

```
bid_sections:
  technical:
  commercial:
  legal:
  pricing:
  eligibility:
  forms:
  approvals:
```

```
Each section:
  status:
  owner:
  deadline:
  blockers:
  completion:
```

Conflict detection examples

```
Technical Agent: 12-month project duration
Commercial Agent: contract requires 9 months
Estimation Agent: priced 10 months

=> Escalate schedule/cost inconsistency

Procurement: vendor quote valid for 30 days
Tender: bid validity is 180 days

=> Pricing exposure detected
```

Decision object

```
decision_id:
topic:
options:
recommended_option:
financial_effect:
risk_effect:
deadline:
decision_owner:
status:
```

Outputs

- Master Bid Tracker
- Agent Conflict Register
- Outstanding Decisions

- Missing Information
- Bid Readiness Score
- Leadership Review Pack
- Consolidated Draft Bid

11. Senior Leadership / Management Agent

ROLE Provide the decision framework for go/no-go, risk appetite, pricing strategy, and final authorization. High-impact choices remain human-approved.

Decision inputs

```
strategic_fit:
technical_capability:
expected_revenue:
expected_margin:
cash_requirement:
risk_exposure:
resource_capacity:
customer_value:
probability_of_win:
```

Decision output

```
decision:
  GO | CONDITIONAL_GO | NO_GO

approved_bid_price:
minimum_acceptable_margin:
approved_contingency:
conditions: []
accepted_risks: []
rejected_risks: []
required_negotiations: []
approver:
```

Human approval required: Final price, major contract deviation acceptance, bid submission, guarantees, bonds, and exceptional risk acceptance must not be auto-committed by the system.

12. Document Control / Compliance Agent

ROLE Maintain continuous compliance with tender submission requirements from the moment requirements are parsed until submission.

Continuous execution: The compliance matrix must run early and continuously, not only at the end. Bid disqualification often comes from missing mandatory evidence, signatures, bonds, forms, or formatting rules.

```
requirement_id:
requirement:
mandatory:
source:
response_owner:
required_document:
document_available:
response_status:
  complete | partial | missing | not_applicable
signature_required:
notarization_required:
expiry_date:
submission_location:
blocking_issue:
```

Responsibilities

- Detect missing mandatory documents and expired certificates.
- Detect unsigned forms, missing bid security, and notarization requirements.
- Track page limits, naming conventions, formatting, and portal upload instructions.
- Detect missing responses and contradictory answers.
- Maintain version register and final submission checklist.

Outputs

- Live Compliance Matrix
- Missing Document List
- Submission Readiness Score
- Final Document Package
- Submission Checklist
- Version Register

13. Company Memory Layer and Match Engine

Company knowledge should be a persistent intelligence layer underneath every tender-specific agent, rather than another one-time agent in the execution chain.

Company Memory contents

- Company profile and industries
- past projects and values
- client references
- technical capabilities
- employee CVs and credentials
- equipment
- certifications
- legal registrations
- insurance
- approved vendors
- historical procurement rates
- historical cost data
- standard contract positions
- risk appetite
- previous tender responses
- financial information and case studies

Company Knowledge object

```
company:
  name:
  countries:
  industries:
  capabilities:
  certifications:
```

```

- name:
  certificate_number:
  issue_date:
  expiry_date:
  document_reference:

projects:
- project_name:
  customer:
  scope:
  value:
  start_date:
  completion_date:
  similarity_tags:
  evidence:

equipment: []
employees: []
standard_rates: []
approved_vendors: []
legal_registrations: []
standard_contract_positions: []
risk_policy: []

```

Company Match Engine

Every extracted tender requirement should be matched automatically against verified company memory. This creates a first-pass compliance and evidence map before specialist drafting begins.

```

Tender requirement:
"Bidder must demonstrate at least three similar projects
above $10M completed during the last five years."

Result:
requirement: REQ-029
required:
  minimum_projects: 3
  minimum_value: 10000000
  period: 5 years
matching_projects:
- Project Alpha
- Project Delta
match_count: 2
status: NON_COMPLIANT
gap: "Only two qualifying projects found."
recommended_action:
"Determine whether JV/consortium experience is permitted."

```

Automatic routing: This gap should feed Document Control, Legal, Bid Manager, and Risk Matrix without waiting for a human to re-enter it.

14. Inter-Agent Contracts and Conflict Handling

Agent message contract

```

message_id:
from_agent:
to_agent:
type:
  information | request | conflict | risk | clarification | decision_required | completed_output
priority:
  low | medium | high | critical
related_requirement_ids: []
payload:
source_references: []
requires_response:
deadline:

```

Conflict object

```

conflict_id:
agents:
- Technical

```

```

- Estimation
issue:
agent_positions:
financial_impact:
schedule_impact:
compliance_impact:
recommended_resolution:
decision_owner:
status:
  open | resolved

```

Assumption object

```

assumption_id:
agent:
assumption:
reason:
financial_impact:
risk_if_wrong:
requires_validation:
validation_owner:
status:
  open | confirmed | rejected

```

Clarification question object

```

clarification_id:
tender_clause:
question:
reason:
related_agents:
impact_if_unanswered:
deadline:
priority:

```

Confidence system

HIGH Direct tender/company evidence exists.

MEDIUM Reasonable interpretation supported by partial evidence.

LOW Insufficient information or an assumption is required.

UNKNOWN No reliable evidence.

Visibility: Low-confidence conclusions should automatically appear in the Bid Manager dashboard.

15. Human Approval Gates and Agent Execution States

Gate 1 - Bid Qualification

Pursue / Do Not Pursue after initial tender analysis.

Gate 2 - Technical & Eligibility Feasibility

Can we realistically comply before significant bid effort?

Gate 3 - Commercial & Legal Risk

Are deviations acceptable?

Gate 4 - Pricing Approval

Approve final price, margin, and contingency.

Gate 5 - Submission Approval

Authorize final bid submission.

Control boundary: TenderMind must not cross Gate 4 or Gate 5 without explicit authorization from an approved human decision-maker.

Agent execution states

```
WAITING
READY
RUNNING
BLOCKED
NEEDS_INPUT
NEEDS_APPROVAL
COMPLETED
FAILED
STALE
```

STALE is critical in a dependency-driven system. If an upstream fact changes, downstream conclusions may no longer be valid.

```
Procurement updates a steel price
    |
    v
Procurement -> COMPLETED
Estimation  -> STALE
Risk Matrix -> STALE
Bid Manager Summary -> STALE
Leadership Pricing Approval -> STALE
```

16. Event-Driven Recalculation and Bid Readiness

Dependency-based update flow

```
Vendor Quote Updated
    |
    v
Procurement Output Updated
    |
    v
Estimation Recalculates
    |
    v
Margin Changes
    |
    v
Risk Exposure Recalculates
    |
    v
Bid Manager Updated
    |
    v
Leadership Approval Invalidated
```

Bid readiness model

- Technical Compliance
- Eligibility Compliance
- Commercial Review Completion
- Legal Review Completion
- Pricing Completion
- Document Completion
- Outstanding Critical Risks
- Outstanding Decisions

```

bid_readiness:
  overall: 78
  technical: 94
  commercial: 82
  legal: 100
  pricing: 87
  documents: 62
  eligibility: 91
  critical_blockers: 2
  status: NOT_READY_FOR_SUBMISSION

```

Important: A high readiness score must never override a mandatory compliance failure.

17. Mandatory Stop Conditions and Prompt Standard

Submission blockers

- Mandatory eligibility requirement = NON_COMPLIANT
- Required bid bond = MISSING
- Required signature = MISSING
- Tender deadline passed
- Critical pricing section incomplete
- Final price not approved
- Critical legal risk awaiting approval
- Mandatory tender form missing
- Required certificate expired
- Unresolved submission-format violation

Agent prompt standard

```

ROLE
OBJECTIVE
AUTHORIZED DATA SOURCES
INPUTS
TASKS
DECISION RULES
OUTPUT SCHEMA
ESCALATION RULES
PROHIBITED ACTIONS
CONFIDENCE RULES
SOURCE CITATION RULES

```

Example specialist instruction

```

You are the TenderMind Legal Agent.

Your job is to identify legal and eligibility risks from tender documents.

You may only use:
1. Tender Workspace
2. Company Knowledge Base
3. Approved company legal policies

Do not invent legal facts.

For every material conclusion provide:
- source document
- page
- section / clause
- risk severity
- confidence
- recommended action

If evidence is insufficient, return UNKNOWN.
Do not make final legally binding decisions.

```

Critical issues must be escalated to the Risk Matrix and Bid Manager.
Return output strictly according to the LegalRisk schema.

18. Final Design Principle

TenderMind should not behave like a generic document generator.

Avoid this

```
Upload Tender
  |
  v
Ask AI
  |
  v
Generate Proposal
```

Build this instead

```
Upload Tender
  |
  v
Understand Tender
  |
  v
Build Requirement Registry
  |
  v
Compare Against Company Memory
  |
  v
Identify Missing Information
  |
  v
Specialist Agent Analysis
  |
  v
Cross-Agent Conflict Detection
  |
  v
Cost + Risk Analysis
  |
  v
Management Decisions
  |
  v
Continuous Compliance Review
  |
  v
Controlled Bid Assembly
  |
  v
Human Approval
  |
  v
Submission
```

Product positioning: TenderMind is not simply AI-generated tender writing. It is a Tender Decision and Bid Intelligence System that recreates the coordination, controls, reasoning, and institutional knowledge of an experienced bid team.